NEET UG BIOLOGY TUTORIAL

Plant Growth and Development

Syllabus Context: Plant Physiology — Growth, Meristems, Growth Curves & Rates, Differentiation, Dedifferentiation, Redifferentiation, Developmental Plasticity, and Plant Growth Regulators (Auxins, Gibberellins, Cytokinins, Ethylene, and Abscisic Acid).

1. Learning Objectives & Chapter Overview

After studying this tutorial, you will be able to:

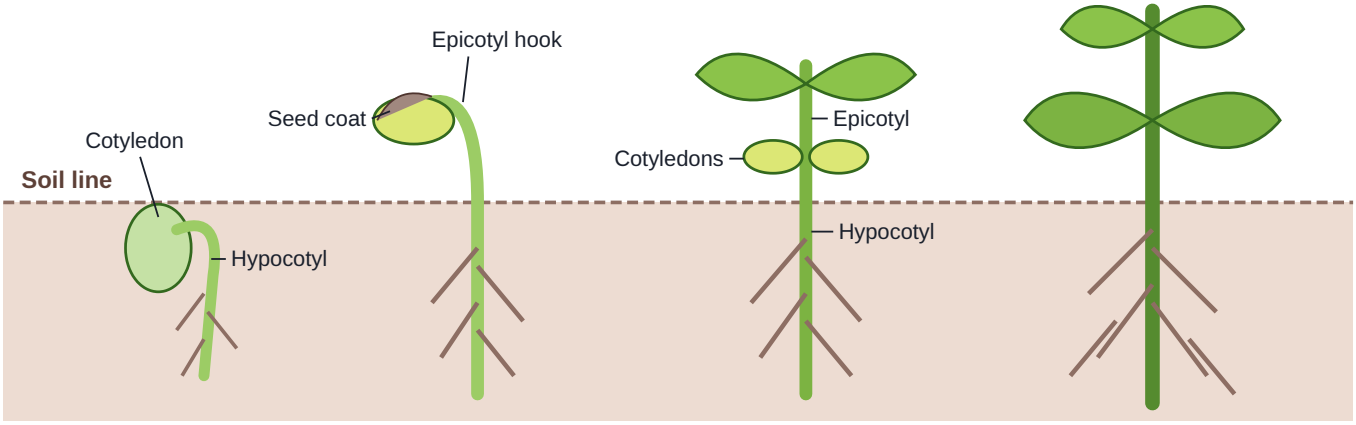
- Define plant growth, understand its indeterminate and open nature, and identify the cellular parameters used to measure it.
- Distinguish between the meristematic, elongation, and maturation phases of growth in a plant axis.
- Analyze arithmetic and geometric growth models mathematically and compare absolute and relative growth rates.
- Explain the processes of differentiation, dedifferentiation, and redifferentiation with anatomical examples.
- Define development and developmental plasticity (heterophylly) in response to life phases and environmental habitats.
- Trace the historical discovery, chemical nature, physiological effects, and agricultural applications of the five major groups of Plant Growth Regulators (PGRs).

Overview of Plant Development

In a flowering plant, structures such as roots, stems, leaves, flowers, fruits, and seeds arise in a precise and orderly sequence. While trees continue to increase in height or girth over a period of time, their leaves, flowers, and fruits have limited dimensions and appear and fall periodically or repeatedly. Similarly, the vegetative phase always precedes flowering in a plant.

All cells of a plant are descendants of a single fertilised egg—the **zygote**. Despite sharing the same genetic origin, these cells acquire diverse structural and functional attributes to form tissues and organs. **Development is the sum of two processes: growth and differentiation.** During development, a complex body organisation is formed that produces roots, leaves, branches, flowers, fruits, and seeds, and eventually the plant dies.

The very first step in the process of plant growth is **seed germination**. A seed germinates when favourable environmental conditions exist. In the absence of favourable conditions, seeds do not germinate and enter a period of suspended growth or rest. Once favourable conditions return, the seeds resume metabolic activities and growth takes place. All these developmental processes are governed by both **intrinsic (internal)** and **extrinsic (external)** factors.



Tutorial Diagram 1: Epigeal seed germination and progressive seedling development in bean, showing the hypocotyl, cotyledons, epicotyl hook, and foliage leaves.

2. Core Concepts: Plant Growth

Growth is regarded as one of the most fundamental and conspicuous characteristics of a living being.

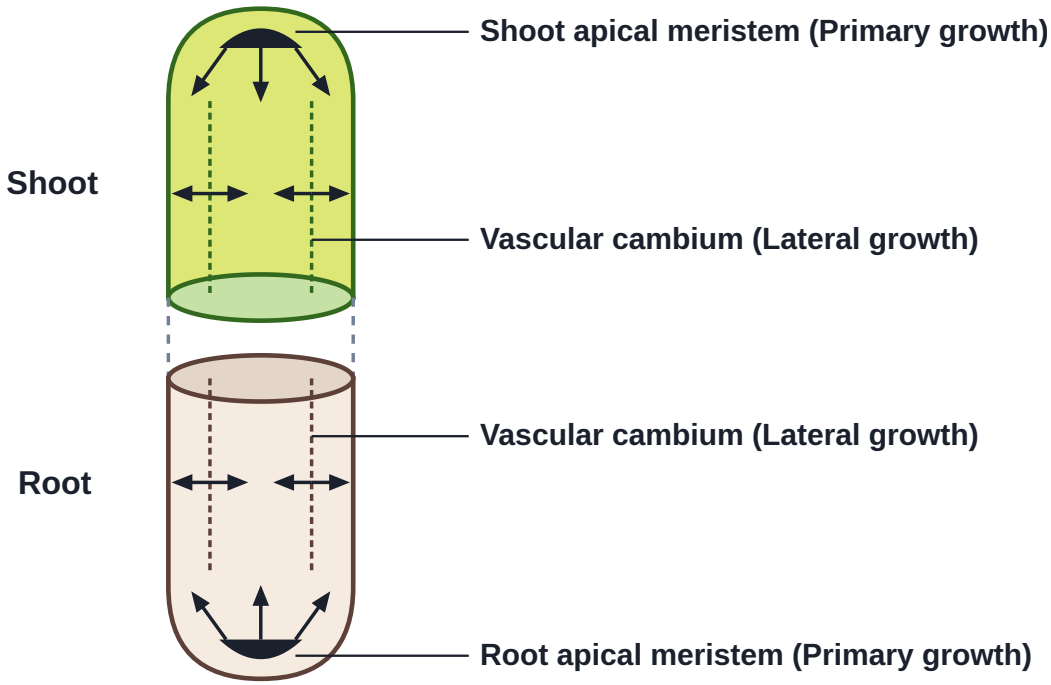
> **Definition of Growth:** Growth is defined as an **irreversible permanent increase in size** of an organ or its parts or even of an individual cell. Generally, growth is accompanied by metabolic processes (both anabolic and catabolic) that occur at the expense of energy.

Conceptual Distinction: The expansion of a leaf is true biological growth because it involves permanent cellular enlargement and metabolic energy expenditure. In contrast, the swelling of a piece of dry wood when placed in water is **imbibition** (a physical, reversible absorption of water without metabolic energy expenditure) and is *not* considered biological growth.

2.1 Plant Growth Generally is Indeterminate

Plants retain the capacity for unlimited growth throughout their life. This unique ability is due to the presence of **meristems** at certain locations in their body. The cells of meristems have the capacity to divide and self-perpetuate. However, the product cells soon lose the capacity to divide and differentiate to make up the plant body.

- **Open Form of Growth:** A form of growth wherein new cells are always being added to the plant body by the activity of the meristem is called the **open form of growth**. If a meristem ever ceases to divide, growth at that point stops (determinate growth, as seen in certain plant organs like leaves, flowers, and fruits).
- **Primary Growth:** The **root apical meristem (RAM)** and the **shoot apical meristem (SAM)** (sometimes along with intercalary meristem) are responsible for the primary growth of plants and principally contribute to the elongation of the plant along its axis.
- **Secondary Growth:** In dicotyledonous plants and gymnosperms, lateral meristems—namely **vascular cambium** and **cork-cambium**—appear later in life. These meristems cause an increase in the **girth** of the organs in which they are active, which is known as secondary growth.



Tutorial Diagram 2: Locations of root apical meristem, shoot apical meristem, and vascular cambium. Arrows indicate the direction of growth of cells and organs.

2.2 Growth is Measurable

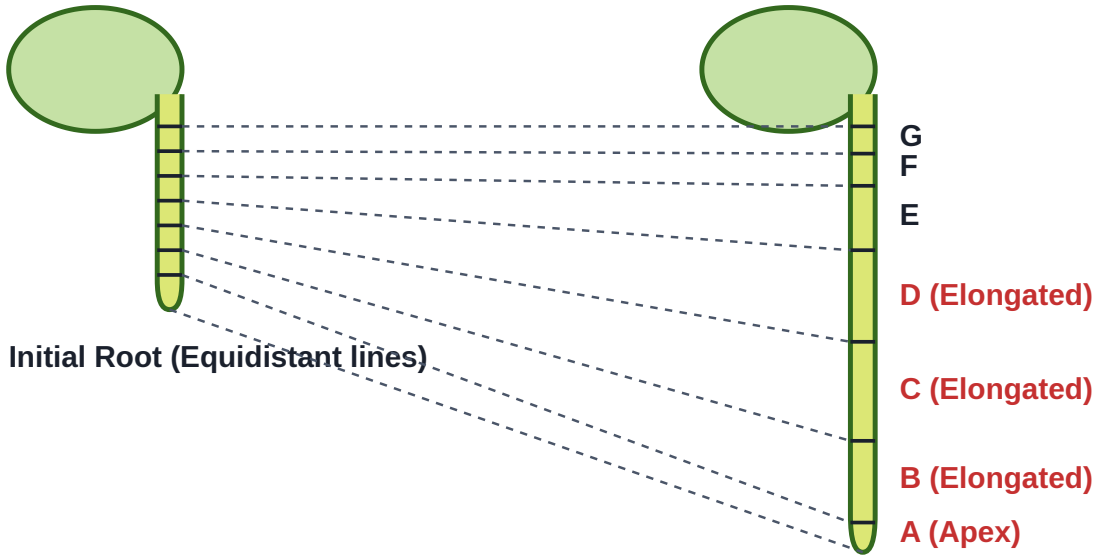
At a cellular level, growth is principally a consequence of an **increase in the amount of protoplasm**. Since an increase in protoplasm is difficult to measure directly, one generally measures some quantity that is more or less proportional to it. No single parameter is sufficient to demonstrate growth across all organs and stages of a plant's life.

Growth Parameter	Biological Example / Application
Increase in Cell Number	A single **maize root apical meristem** can give rise to more than **17,500 new cells per hour**.
Increase in Cell Size / Volume	Cells in a **watermelon** may increase in size by up to **3,50,000 times**.
Increase in Length	Growth of a **pollen tube** or elongation of a root/stem axis is measured in terms of its length.
Increase in Surface Area	Growth in a flat organ like a **dorsiventral leaf** is measured by increase in surface area.
Fresh Weight & Dry Weight	Used to measure overall biomass accumulation in whole plants, organs, or tissue cultures.

2.3 Phases of Growth

The period of growth in a plant axis (such as a root tip) is divided into three distinct phases:

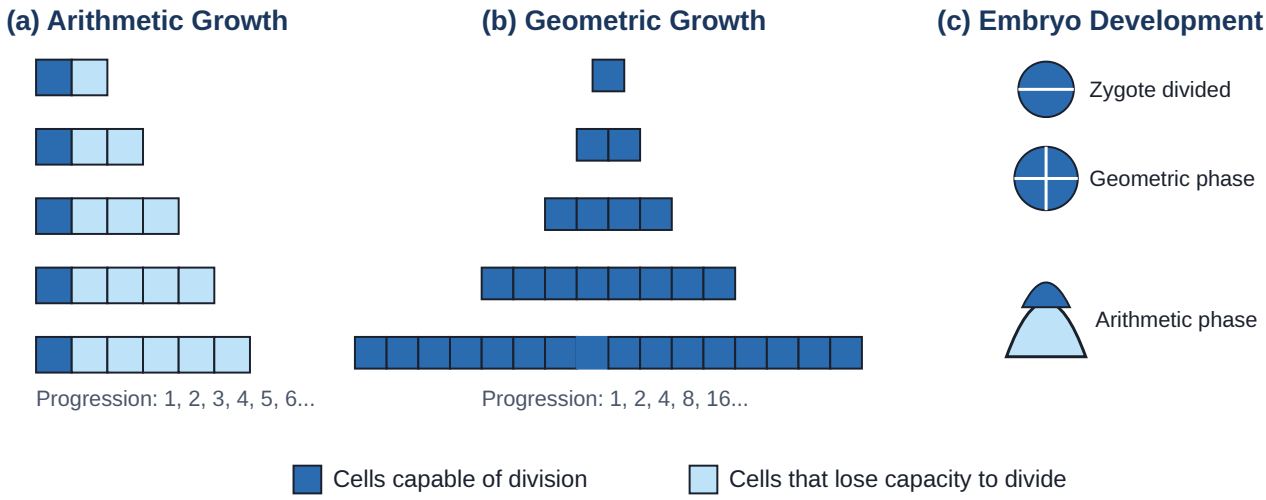
1. **Meristematic Phase:** Represented by the constantly dividing cells at both the root apex and the shoot apex.
   - Cells are **rich in protoplasm** and possess **large, conspicuous nuclei**.
   - Cell walls are **primary in nature, thin, and cellulosic** with **abundant plasmodesmatal connections**.
2. **Phase of Elongation:** Represented by the cells **proximal** (just next, away from the tip) to the meristematic zone.
   - Characterised by **increased vacuolation**, **cell enlargement**, and **new cell wall deposition**.
3. **Phase of Maturation:** Lies further away from the apex, i.e., more proximal to the phase of elongation.
   - Cells attain their **maximal size** in terms of **wall thickening** and **protoplasmic modifications**. Most differentiated plant tissues and cell types represent this phase.



Tutorial Diagram 3: Detection of zones of elongation by the parallel line technique. Zones A, B, C, and D immediately behind the apex have elongated the most.

3. Growth Rates, Mathematical Models & Growth Curves

The increased growth per unit time is termed as **growth rate**. Rate of growth can be expressed mathematically. An organism, or a part of an organism, can produce more cells in a variety of ways, showing either **arithmetic** or **geometrical** growth.



Tutorial Diagram 4: Diagrammatic representation of (a) Arithmetic growth, (b) Geometric growth, and (c) Stages during embryo development showing an initial geometric phase followed by an arithmetic phase.

3.1 Arithmetic Growth

In arithmetic growth, following mitotic cell division, **only one daughter cell continues to divide while the other differentiates and matures**. The simplest expression of arithmetic growth is exemplified by a root elongating at a constant rate. On plotting the length of the organ against time, a **linear curve** is obtained.

$$L_t = L_0 + rt$$

Where L_t = length at time t; L_0 = length at time zero; r = growth rate / elongation per unit time.

3.2 Geometrical Growth & Sigmoid Curve

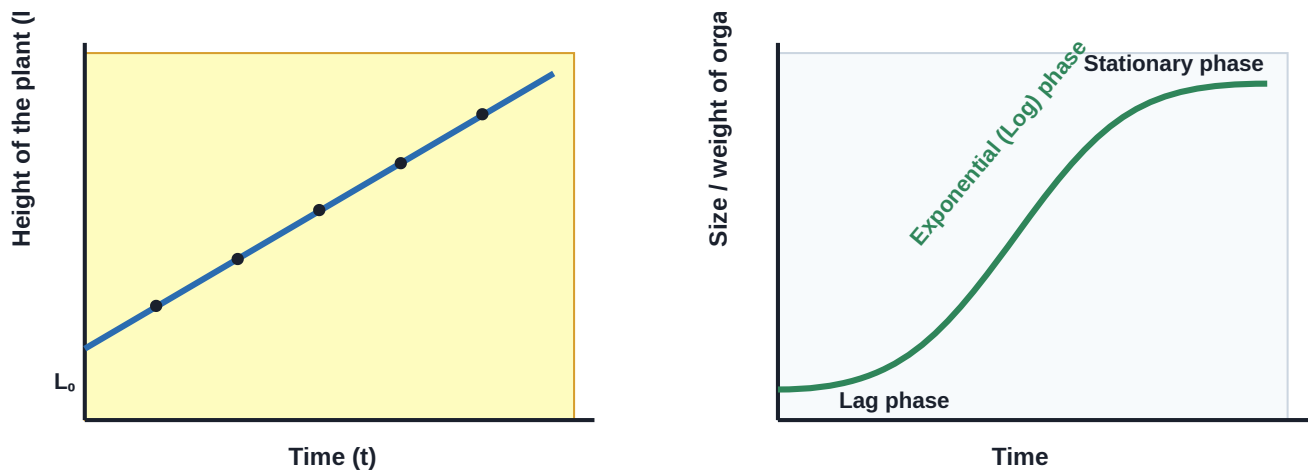
In geometrical growth, **both the progeny cells following mitotic cell division retain the ability to divide and continue to do so**. In most biological systems, geometrical growth exhibits three distinct phases:

- **Lag Phase:** Initial growth is slow.
- **Log or Exponential Phase:** Growth increases rapidly thereafter at an exponential rate.
- **Stationary Phase:** With limited nutrient supply, growth slows down and reaches a plateau (in the context of the whole life cycle, this is also referred to as the senescent phase).

If we plot the parameter of growth against time, we get a typical **sigmoid or S-curve**. A sigmoid curve is a characteristic of living organisms growing in a natural environment and is typical for all cells, tissues, and organs of a plant (as well as cells in culture and trees showing seasonal activities).

$$W_1 = W_0 e^{rt}$$

Where W_1 = final size (weight, height, number, etc.); W_0 = initial size at the beginning of the period; r = relative growth rate (also called **efficiency index**, a measure of the ability of the plant to produce new plant material); t = time of growth; e = base of natural logarithms. Note that the final size W_1 depends on the initial size W_0.



Tutorial Diagram 5: Left — Constant linear growth curve obtained in arithmetic growth. Right — Idealised sigmoid (S-shaped) growth curve typical of geometric growth in natural environments.

3.3 Quantitative Comparisons: Absolute vs. Relative Growth Rate

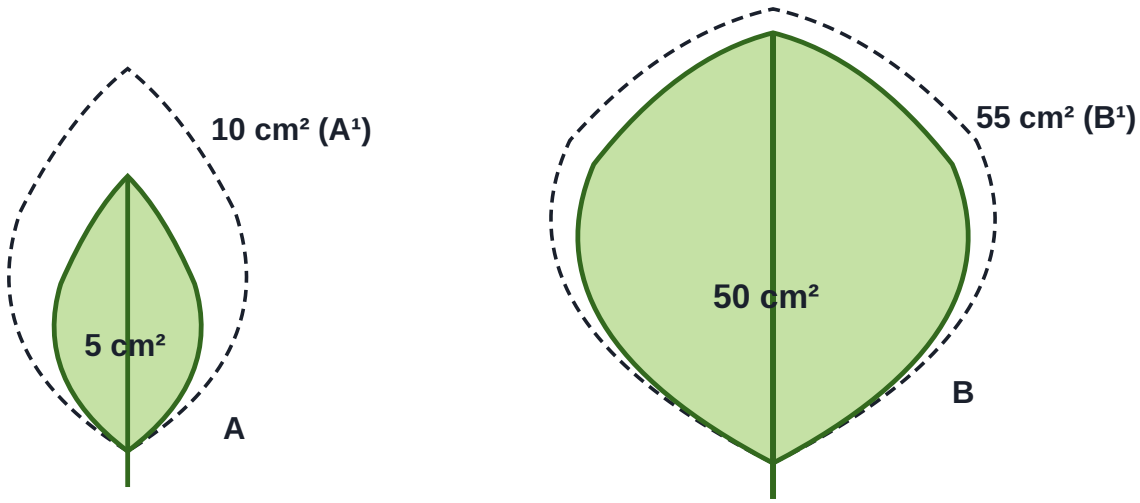
Quantitative comparisons between the growth of living systems can be made in two ways:

1. **Absolute Growth Rate (AGR):** The measurement and comparison of total growth per unit time.

$$\text{AGR} = \frac{\text{Final Parameter} - \text{Initial Parameter}}{\text{Time}}$$

2. **Relative Growth Rate (RGR):** The growth of the given system per unit time expressed on a common basis, e.g., per unit initial parameter.

$$\text{RGR} = \frac{\text{Absolute Growth Rate}}{\text{Initial Parameter}}$$



Tutorial Diagram 6: Comparison of absolute and relative growth rates. Both leaves A and B have increased their area by 5 cm² in a given time to produce A¹ and B¹ leaves, respectively.

3.4 Conditions for Growth

Essential elements required for plant growth include:

- **Water:** Plant cells grow in size by cell enlargement, which requires water. **Turgidity** of cells helps in extension growth. Water also provides the medium for enzymatic activities needed for growth.
- **Oxygen:** Helps in releasing **metabolic energy** essential for growth activities.
- **Nutrients:** Macro and micro essential elements are required by plants for the **synthesis of protoplasm** and act as a source of energy.
- **Temperature:** Every plant has an **optimum temperature range** best suited for its growth; any deviation from this range could be detrimental to its survival.
- **Environmental Signals:** Factors such as **light and gravity** affect certain phases and stages of growth.

4. Differentiation, Dedifferentiation, Redifferentiation & Development

4.1 Cellular Transitions in Plants

Process	Definition	Key Examples from Plant Anatomy & Physiology
Differentiation	The act leading to maturation where cells derived from root/shoot apical meristems and cambium mature to perform specific functions. Cells undergo structural changes in both cell walls and protoplasm.	To form a **tracheary element**, cells lose their protoplasm and develop strong, elastic, **lignocellulosic secondary cell walls** to carry water to long distances even under extreme tension.
Dedifferentiation	The phenomenon wherein living differentiated cells, which had lost the capacity to divide, **regain the capacity of division** under certain conditions.	Formation of secondary meristems—**interfascicular cambium** and **cork cambium (phellogen)**—from fully differentiated parenchyma cells; also parenchyma cells dividing in plant tissue culture to form callus.
Redifferentiation	Cells produced by dedifferentiated meristems divide and once again **lose the capacity to divide**, maturing to perform specific functions.	In a woody dicotyledonous plant: **secondary xylem, secondary phloem, cork (phellem), and secondary cortex (phelloderm)**.

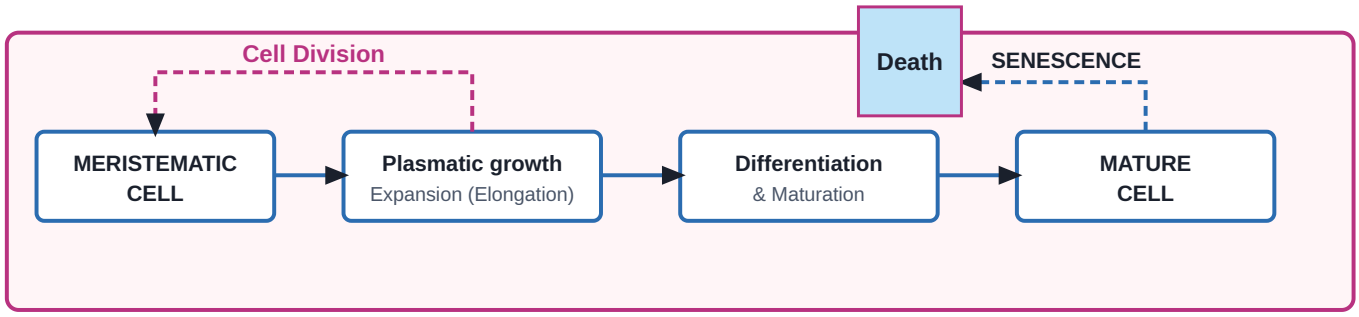
Why Differentiation in Plants is also "Open":

Just as growth in plants is open (indeterminate or determinate), **differentiation in plants is also open** because cells or tissues arising out of the same meristem have different structures at maturity. The final structure at maturity of a cell or tissue is determined by the **location of the cell within**. For example, cells

positioned away from root apical meristems differentiate as **root-cap cells**, while those pushed to the periphery mature as **epidermis**.

4.2 Development and Plasticity

Development is a term that includes all changes that an organism goes through during its life cycle from the **germination of the seed to senescence**. It applies equally to cells, tissues, and organs.

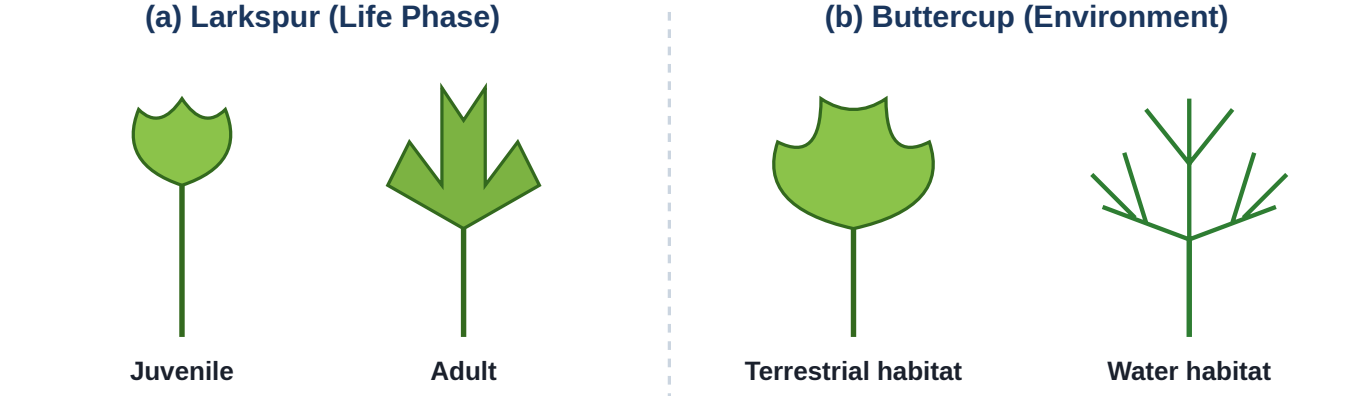


Tutorial Diagram 7: Sequence of the developmental process in a plant cell (also applicable to tissues and organs).

Developmental Plasticity (Heterophylly)

Plants follow different pathways in response to their environment or phases of life to form different kinds of structures. This ability is called **plasticity**. A classic example of plasticity is **heterophylly** (presence of more than one type of leaf on the same plant):

- **Heterophylly due to Phases of Life (Intrinsic / Developmental):** Seen in **cotton, coriander, and larkspur**. In such plants, the leaves of the **juvenile plant** are different in shape from those in **mature (adult) plants**.
- **Heterophylly due to Environment (Extrinsic / Environmental):** Seen in **buttercup** (*Ranunculus*). There is a marked difference in the shapes of leaves produced in **air (terrestrial habitat)** compared to dissected leaves produced in **water (aquatic habitat)**.



Tutorial Diagram 8: Heterophylly representing plasticity in (a) Larkspur (juvenile vs. adult leaf shapes) and (b) Buttercup (terrestrial vs. water habitat leaf shapes).

Factors Controlling Development

Development in plants (both growth and differentiation) is controlled by:

- **Intrinsic Factors:** Include both **intracellular (genetic/genomic)** factors and **intercellular factors** (chemical substances such as Plant Growth Regulators).
- **Extrinsic Factors:** Include **light, temperature, water, oxygen, nutrition**, etc.

5. Plant Growth Regulators (PGRs)

Plant Growth Regulators (variously called plant growth substances, plant hormones, or phytohormones) are small, simple molecules of diverse chemical composition.

5.1 Chemical Nature & Functional Classification of PGRs

PGR Group	Chemical Composition / Precursor Derivative	Functional Category
Auxins	Indole compounds (e.g., **indole-3-acetic acid, IAA**)	Plant Growth Promoter
Cytokinins	Adenine derivatives (e.g., **N^6-furfurylamino purine, kinetin**)	Plant Growth Promoter
Gibberellins	Terpenes (e.g., **gibberellic acid, GA_3**)	Plant Growth Promoter
Abscisic Acid (ABA)	Derivatives of carotenoids	Plant Growth Inhibitor (Wounds, biotic/abiotic stresses, dormancy, abscission)
Ethylene	Simple gases (**C_2H_4**)	Fits either group, but is **largely an inhibitor** of growth activities

5.2 Accidental Discovery of the Five Major PGR Groups

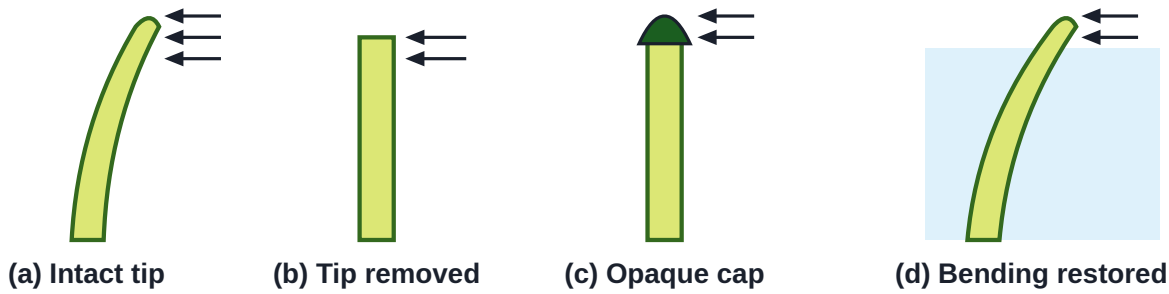
1. **Auxins:**
   - **Charles Darwin and his son Francis Darwin** observed that the coleoptiles of **canary grass** responded to unilateral illumination by growing towards the light source (**phototropism**). They concluded that the tip of the coleoptile was the site of a transmittable influence that caused the bending of the entire coleoptile.
   - **F.W. Went** later isolated auxin from the tips of coleoptiles of **oat seedlings**.

2. **Gibberellins:**
   - The **'bakanae' (foolish seedling) disease** of rice seedlings was caused by a fungal pathogen, ***Gibberella fujikuroi***.
   - **E. Kurosawa (1926)** reported the appearance of symptoms of the disease in uninfected rice seedlings when they were treated with sterile filtrates of the fungus. The active substances were later identified as **gibberellic acid**.

3. **Cytokinins:**
   - **F. Skoog and his co-workers** observed that from the internodal segments of **tobacco stems**, the **callus** (a mass of undifferentiated cells) proliferated only if, in addition to auxins, the nutrients medium was supplemented with one of the following: extracts of vascular tissues, yeast extract, coconut milk, or DNA.
   - **Miller et al. (1955)** later identified and crystallised the cytokinesis-promoting active substance from autoclaved herring sperm DNA and termed it **kinetin**.

4. **Abscisic Acid (ABA):**
   - During the mid-1960s, three independent researchers reported the purification and chemical characterisation of three different kinds of inhibitors: **inhibitor-B, abscission II, and dormin**.
   - Later, all three were proved to be chemically identical and named **abscisic acid (ABA)**.

5. **Ethylene:**
   - **H.H. Cousins (1910)** confirmed the release of a volatile substance from **ripened oranges** that hastened the ripening of stored **unripened bananas**. Later, this volatile substance was identified as **ethylene**.

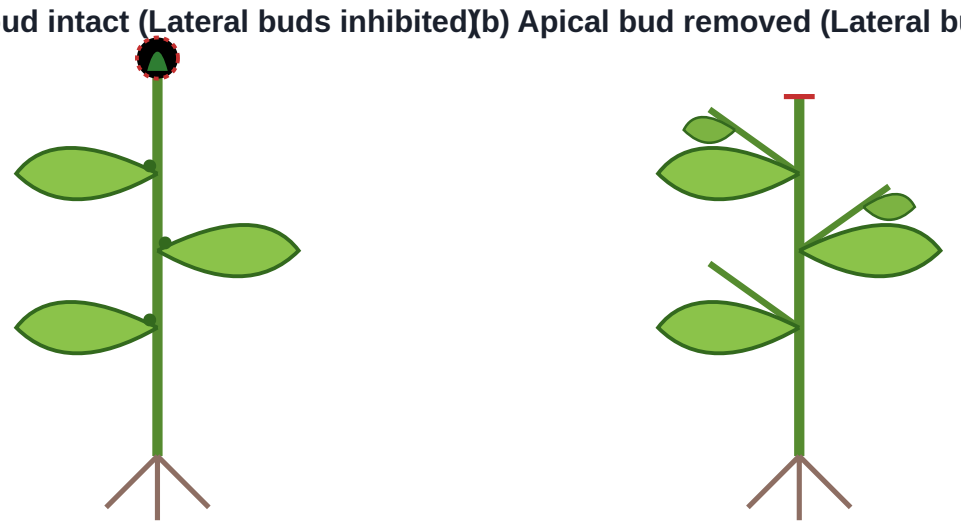


Tutorial Diagram 9: Experiments demonstrating that the tip of the coleoptile is the source of the transmittable influence (auxin) responsible for phototropic bending towards unilateral illumination (arrows).

5.3 Physiological Effects & Applications of PGRs

1. Auxins

- **Origin & Isolation:** From Greek *'auxein'* (to grow); first isolated from **human urine**. Produced generally by growing apices of stems and roots, from where they migrate to regions of action.
- **Natural vs. Synthetic Auxins:**
  - **Natural Auxins (isolated from plants): IAA** (indole-3-acetic acid) and **IBA** (indole butyric acid).
  - **Synthetic Auxins: NAA** (naphthalene acetic acid) and **2, 4-D** (2, 4-dichlorophenoxyacetic acid).
- **Physiological Roles & Applications:**
  - **Rooting:** Help to initiate rooting in stem cuttings (widely used for plant propagation).
  - **Flowering:** Promote flowering, e.g., in **pineapples**.
  - **Abscission Control:** Prevent fruit and leaf drop at **early stages**, but promote the abscission of **older mature leaves and fruits**.
  - **Apical Dominance:** In most higher plants, the growing apical bud inhibits the growth of lateral (axillary) buds. Removal of shoot tips (**decapitation**) results in the growth of lateral buds. Widely applied in **tea plantations and hedge-making** (to induce bushy lateral branching and more pluckable shoots).
  - **Parthenocarpy:** Induce seedless fruit development, e.g., in **tomatoes**.
  - **Weedicide / Herbicide: 2, 4-D** is widely used to kill **dicotyledonous weeds** and does not affect mature monocotyledonous plants; used to prepare weed-free lawns by gardeners.
  - **Vascular & Cellular Role:** Control **xylem differentiation** and help in cell division.



Tutorial Diagram 10: Apical dominance in plants — (a) Intact apical bud suppresses axillary buds; (b) Removal of shoot tip (decapitation) releases lateral buds to grow into branches.

2. Gibberellins

- **Nature & Diversity:** Promotory PGRs; more than 100 gibberellins are reported from widely different organisms such as fungi and higher plants, denoted as GA_1, GA_2, GA_3, etc. **Gibberellic acid (GA_3)** was one of the first discovered and remains the most intensively studied form. **All GAs are acidic.**
- **Physiological Roles & Applications:**
  - **Axis Elongation in Fruits:** Increase the length of **grape stalks**; cause fruits like **apple to elongate and improve its shape**.
  - **Senescence Delay:** Delay senescence so fruits can be left on the tree longer to extend the market period.
  - **Brewing Industry:** GA_3 is used to speed up the **malting process** in the brewing industry.
  - **Sugarcane Yield:** Sugarcane stores carbohydrate as sugar in its stems. Spraying sugarcane crop with gibberellins increases stem length, increasing the yield by as much as **20 tonnes per acre**.
  - **Early Maturity in Conifers:** Spraying **juvenile conifers** with GAs hastens the maturity period, leading to **early seed production**.
  - **Bolting:** Promote **bolting** (internode elongation just prior to flowering) in **beet, cabbages**, and many plants with a rosette habit.

3. Cytokinins

- **Discovery & Synthesis:** Have specific effects on cytokinesis. Discovered as **kinetin** (a modified form of adenine, a purine) from autoclaved herring sperm DNA. **Kinetin does not occur naturally in plants.** Search for natural substances led to the isolation of **zeatin** from **corn-kernels and coconut milk**. Natural cytokinins are synthesised in regions where rapid cell division occurs: **root apices, developing shoot buds, young fruits**, etc.
- **Physiological Roles & Applications:**
  - Help to produce **new leaves, chloroplasts in leaves, lateral shoot growth, and adventitious shoot formation**.
  - Help **overcome apical dominance** (antagonistic to auxin in lateral bud growth).
  - Promote **nutrient mobilisation**, which helps in the **delay of leaf senescence**.

4. Ethylene

- **Synthesis:** A simple gaseous PGR synthesised in large amounts by **tissues undergoing senescence and ripening fruits**.

- **Physiological Roles & Applications:**
  - **Triple Response in Seedlings:** Causes horizontal growth of seedlings, swelling of the axis, and **apical hook formation in dicot seedlings**.
  - **Senescence & Abscission:** Promotes senescence and abscission of plant organs, especially of leaves and flowers.
  - **Fruit Ripening & Respiratory Climactic:** Highly effective in fruit ripening; enhances the respiration rate during ripening of fruits, a rise called **respiratory climactic**.
  - **Breaking Dormancy:** Breaks seed and bud dormancy; initiates **germination in peanut seeds** and **sprouting of potato tubers**.
  - **Submergence Escape:** Promotes **rapid internode/petiole elongation in deep water rice plants**, helping leaves and upper parts of the shoot remain above water.
  - **Absorption Surface:** Promotes **root growth and root hair formation**, helping plants increase their absorption surface.
  - **Flowering & Fruit-Set:** Initiates flowering and **synchronises fruit-set in pineapples**; induces **flowering in mango**.
  - **Ethephon Applications:** The most widely used compound as a source of ethylene is **ethephon** (readily absorbed in aqueous solution and releases ethylene slowly). Ethephon hastens fruit ripening in **tomatoes and apples**, accelerates abscission in flowers and fruits (**thinning of cotton, cherry, walnut**), and promotes **female flowers in cucumbers**, thereby increasing the yield.

5. Abscisic Acid (ABA)

- **General Inhibitor:** Acts as a general plant growth inhibitor and an inhibitor of plant metabolism.
- **Stress Hormone:** Stimulates the **closure of stomata** and increases the tolerance of plants to various kinds of stresses; therefore, it is called the **stress hormone**.
- **Dormancy & Seed Maturation:** Inhibits seed germination; plays an important role in **seed development, maturation, and dormancy**. By inducing dormancy, ABA helps seeds withstand **desiccation** and other factors unfavourable for growth.
- **Antagonism:** In most situations, **ABA acts as an antagonist to Gibberellins (GAs)**.

5.4 PGR Interactions & Extrinsic Controls

For every phase of growth, differentiation, and development of plants, one or the other PGR has a role to play. These roles can be **complementary or antagonistic**, and **individualistic or synergistic**. Multiple PGRs interact to affect events such as:

- **Dormancy in seeds/buds:** ABA induces dormancy while Gibberellins and Ethylene break dormancy.
- **Apical dominance:** Auxin promotes apical dominance (inhibits lateral buds) while Cytokinin overcomes apical dominance (promotes lateral buds).
- **Senescence & Abscission:** Ethylene and ABA promote senescence/abscission, whereas Auxin (at early stages), Gibberellins, and Cytokinins delay senescence.

Remember that PGRs represent only one kind of intrinsic control. Along with **genomic control** and **extrinsic factors** (particularly temperature and light), they regulate events such as vernalisation, flowering (photoperiodism), dormancy, seed germination, and plant movements.

6. Worked Examples & Conceptual Applications

> #### Worked Example 1: Quantitative Analysis of Absolute and Relative Growth Rates
>
> **Problem Statement:** Two leaves, Leaf A and Leaf B, have initial surface areas of $5\,\mathrm{cm}^2$ and $50\,\mathrm{cm}^2$, respectively. Over a period of 1 unit time, both leaves expand by $5\,\mathrm{cm}^2$ to reach final areas of $A^1 = 10\,\mathrm{cm}^2$ and $B^1 = 55\,\mathrm{cm}^2$. Calculate their Absolute Growth Rates (AGR) and Relative Growth Rates (RGR) and identify which leaf grows faster on a relative basis.

Solution:

- **Step 1: Calculate Absolute Growth Rate (AGR)**

$$\text{AGR of Leaf A} = 10\,\text{cm}^2 - 5\,\text{cm}^2 = 5\,\text{cm}^2 \text{ per unit time}$$
$$\text{AGR of Leaf B} = 55\,\text{cm}^2 - 50\,\text{cm}^2 = 5\,\text{cm}^2 \text{ per unit time}$$

  Both leaves have the exact same Absolute Growth Rate of $5\,\text{cm}^2$ per unit time.

- **Step 2: Calculate Relative Growth Rate (RGR)**

$$\text{RGR of Leaf A} = \frac{5\,\text{cm}^2}{5\,\text{cm}^2} \times 100 = 100\% \text{ (or 1.0 per unit time)}$$
$$\text{RGR of Leaf B} = \frac{5\,\text{cm}^2}{50\,\text{cm}^2} \times 100 = 10\% \text{ (or 0.1 per unit time)}$$

- **Conclusion: Leaf A** exhibits a much higher Relative Growth Rate (100% vs. 10%) because its initial parameter ($5\,\text{cm}^2$) was ten times smaller than that of Leaf B ($50\,\text{cm}^2$).

Worked Example 2: Selecting the Right PGR for Horticultural Tasks

Problem Statement: Which specific Plant Growth Regulator should be used to achieve each of the following physiological outcomes?

a. Induce rooting in a stem twig
b. Quickly ripen a fruit
c. Delay leaf senescence by nutrient mobilisation
d. Induce growth in axillary buds by overcoming apical dominance
e. 'Bolt' a rosette plant (like cabbage or beet)
f. Induce immediate stomatal closure in leaves under water stress

Solution:

(a) **Auxin** (such as IAA, IBA, or NAA) | (b) **Ethylene** (or Ethephon) | (c) **Cytokinin** | (d) **Cytokinin** | (e) **Gibberellin** (GA_3) | (f) **Abscisic acid (ABA)**.

Worked Example 3: Predicting Physiological Consequences

Problem Statement: Predict what would happen in each of the following experimental scenarios:

a. GA_3 is applied to rice seedlings.
b. Dividing cells stop differentiating.
c. A rotten/ripened fruit gets mixed with unripe fruits.
d. You forget to add cytokinin to a tobacco stem callus culture medium containing only auxin.
e. A plant is completely defoliated and exposed to a photoperiodic cycle.

Solution:

- **(a)** The rice seedlings will undergo rapid internode elongation, becoming abnormally tall, thin, and pale—mimicking the symptoms of the 'bakanae' (foolish seedling) disease.
- **(b)** The cells will continue dividing without maturing into specialised tissues, resulting in an undifferentiated mass of cells (such as a **callus** in tissue culture or a **tumour**).
- **(c)** The ripened/rotten fruit releases the volatile gas **ethylene**, which stimulates the respiratory climactic and hastens the ripening of the surrounding unripe fruits.
- **(d)** In the absence of cytokinin (supplemented alongside auxin), the callus will fail to proliferate and shoot bud differentiation will not occur.

- **(e)** A completely defoliated plant will **not** respond to the photoperiodic cycle because **leaves** are the sites of perception of light/dark duration for flowering. (Note also that a short-day plant and a long-day plant can flower simultaneously in a given place if the natural day length lies below the critical photoperiod of the short-day plant and above the critical photoperiod of the long-day plant).

7. NEET UG Practice Questions with Worked Solutions

Note: All questions below are original practice items designed strictly in accordance with the NEET UG examination pattern and the chapter source material.

Practice Question 1 (Single-Choice MCQ)

Which of the following characteristics correctly describes the cells present in the meristematic phase of growth at the root apex?

(A) Large central vacuoles, thick lignified secondary cell walls, and sparse protoplasm

(B) Rich in protoplasm, large conspicuous nuclei, and thin cellulosic primary walls with abundant plasmodesmata

(C) Loss of protoplasm, maximal wall thickening, and absence of plasmodesmatal connections

(D) High vacuolation, rapid cell enlargement, and suberised cell walls

Correct Answer: (B)

Worked Explanation: The constantly dividing cells at the root apex and shoot apex represent the meristematic phase of growth. These cells are rich in protoplasm, possess large conspicuous nuclei, and have thin, primary, cellulosic cell walls with abundant plasmodesmatal connections. Increased vacuolation and cell enlargement occur in the phase of elongation, while maximal wall thickening and protoplasmic modifications occur in the phase of maturation.

Practice Question 2 (Single-Choice MCQ — Mathematical Model)

A root elongating at a constant rate shows arithmetic growth represented by $L_t = L_0 + rt$. If the initial length of a young root is $2.5\ \text{cm}$ and it elongates at a steady rate of $0.4\ \text{cm/day}$, what will be its length after 10 days?

(A) $4.0\ \text{cm}$

(B) $6.5\ \text{cm}$

(C) $10.0\ \text{cm}$

(D) $12.5\ \text{cm}$

Correct Answer: (B)

Worked Explanation: Using the arithmetic growth formula $L_t = L_0 + rt$:
Given $L_0 = 2.5\ \text{cm}$, $r = 0.4\ \text{cm/day}$, and $t = 10\ \text{days}$.

$$L_{10} = 2.5 + (0.4 \times 10) = 2.5 + 4.0 = 6.5\ \text{cm}$$

Practice Question 3 (Statement-Based MCQ)

Read the following two statements carefully and choose the correct option:

Statement I: One single maize root apical meristem can give rise to more than 17,500 new cells per hour, where growth is expressed as an increase in cell number.
Statement II: Cells in a watermelon may increase in size by up to 3,50,000 times, where growth is expressed as an increase in size of the cell.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct, but Statement II is incorrect.

(D) Statement I is incorrect, but Statement II is correct.

Correct Answer: (A)

Worked Explanation: Both factual values are directly established in plant growth measurement: a single maize root apical meristem produces over 17,500 new cells per hour (growth measured by cell number), whereas watermelon cells enlarge up to 3,50,000 times their initial volume (growth measured by cell size).

Practice Question 4 (Single-Choice MCQ)

In the exponential growth equation $W_1 = W_0 e^{rt}$, the parameter r represents:

(A) Absolute growth rate and base of natural logarithms

(B) Relative growth rate and efficiency index of the plant to produce new plant material

(C) Initial size of the organ during the stationary phase

(D) Linear elongation rate during arithmetic cell division

Correct Answer: (B)

Worked Explanation: In $W_1 = W_0 e^{rt}$, r is the relative growth rate and is also the measure of the ability of the plant to produce new plant material, referred to as the **efficiency index**.

Practice Question 5 (Single-Choice MCQ)

The formation of interfascicular cambium and cork cambium from fully differentiated parenchyma cells in a dicot stem is an example of:

(A) Differentiation

(B) Dedifferentiation

(C) Redifferentiation

(D) Determinate growth

Correct Answer: (B)

Worked Explanation: Living differentiated cells (such as medullary ray parenchyma or cortical parenchyma) that have lost the capacity to divide can regain the capacity of division under certain conditions. This phenomenon is termed **dedifferentiation**, and it gives rise to secondary meristems like interfascicular cambium and cork cambium.

Practice Question 6 (Single-Choice MCQ)

Heterophylly in buttercup (*Ranunculus*) differs from heterophylly in cotton, coriander, and larkspur because in buttercup:

(A) Heterophylly is caused by transition from juvenile phase to adult phase.

(B) Heterophylly is an example of plasticity induced by different environments (air vs. water habitat).

(C) Leaves are completely absent in the aquatic habitat.

(D) It is caused by infection with the fungus *Gibberella fujikuroi*.

Correct Answer: (B)

Worked Explanation: In cotton, coriander, and larkspur, the leaves of the juvenile plant differ in shape from those in mature plants (phase-dependent plasticity). In contrast, buttercup exhibits different leaf shapes in air (terrestrial habitat) versus water (aquatic habitat), representing heterophyllous development due to environment.

Practice Question 7 (Matching Type MCQ)

Match the Plant Growth Regulators in Column I with their chemical nature in Column II:

Column I (PGR)	Column II (Chemical Nature)
(P) Indole-3-acetic acid (IAA)	(i) Terpenes
(Q) Kinetin	(ii) Derivatives of carotenoids
(R) Abscisic acid (ABA)	(iii) Indole compounds
(S) Gibberellic acid (GA_3)	(iv) Adenine derivatives (N^6-furfurylamino purine)

(A) P-(iii), Q-(iv), R-(ii), S-(i)

(B) P-(iv), Q-(iii), R-(i), S-(ii)

(C) P-(iii), Q-(ii), R-(iv), S-(i)

(D) P-(i), Q-(iv), R-(ii), S-(iii)

Correct Answer: (A)

Worked Explanation: IAA is an indole compound (iii); Kinetin is an adenine derivative, N^6-furfurylamino purine (iv); Abscisic acid is a derivative of carotenoids (ii); and Gibberellic acid (GA_3) is a terpene (i).

Practice Question 8 (Matching Type MCQ — Discovery of PGRs)

Match the scientists in Column I with their experimental organism/material in Column II:

Column I (Scientist)	Column II (Experimental Material)
(P) Charles and Francis Darwin	(i) Ripened oranges and unripened bananas
(Q) F.W. Went	(ii) Sterile filtrates of *Gibberella fujikuroi* on rice
(R) E. Kurosawa	(iii) Coleoptile tips of oat seedlings
(S) H.H. Cousins	(iv) Coleoptiles of canary grass

(A) P-(iii), Q-(iv), R-(ii), S-(i)

(B) P-(iv), Q-(iii), R-(ii), S-(i)

(C) P-(iv), Q-(ii), R-(iii), S-(i)

(D) P-(ii), Q-(iii), R-(iv), S-(i)

Correct Answer: (B)

Worked Explanation: Charles and Francis Darwin studied phototropism in coleoptiles of canary grass (iv). F.W. Went isolated auxin from tips of coleoptiles of oat seedlings (iii). E. Kurosawa (1926) used sterile filtrates of the fungus *Gibberella fujikuroi* on rice seedlings (ii). H.H. Cousins (1910) studied the volatile substance from ripened oranges affecting stored bananas (i).

Practice Question 9 (Assertion-Reason MCQ)

Assertion (A): Decapitation (removal of shoot tips) is widely practised in tea plantations and hedge-making.
Reason (R): Removal of the growing apical bud releases the lateral (axillary) buds from apical dominance, resulting in dense branching.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Explanation: In most higher plants, the growing apical bud synthesises auxin that inhibits the growth of lateral (axillary) buds—a phenomenon called apical dominance. Removing the shoot tip (decapitation) eliminates the source of apical dominance, allowing lateral buds to sprout into branches. This is directly applied in tea plantations and hedge-making.

Practice Question 10 (Single-Choice MCQ)

Which of the following pairs consists exclusively of synthetic auxins?

(A) IAA and IBA

(B) NAA and 2, 4-D

(C) IAA and NAA

(D) IBA and Zeatin

Correct Answer: (B)

Worked Explanation: IAA (indole-3-acetic acid) and IBA (indole butyric acid) have been isolated from plants (natural auxins). NAA (naphthalene acetic acid) and 2, 4-D (2, 4-dichlorophenoxyacetic acid) are synthetic auxins. Zeatin is a natural cytokinin.

Practice Question 11 (Single-Choice MCQ)

Spraying sugarcane crop with gibberellins increases the yield by as much as 20 tonnes per acre because gibberellins:

(A) Increase the rate of photosynthesis in leaves and induce parthenocarpy

(B) Increase the length of the stem where sugarcane stores carbohydrate as sugar

(C) Promote root hair formation and rapid petiole elongation

(D) Prevent abscission of older mature leaves and kill dicot weeds

Correct Answer: (B)

Worked Explanation: Sugarcane stores carbohydrate as sugar in its stems. Spraying the sugarcane crop with gibberellins increases the length of the stem (internodal elongation), thereby increasing sugar yield by up to 20 tonnes per acre.

Practice Question 12 (Multi-Statement MCQ)

Identify the correct statements regarding Cytokinins:

i. Kinetin was discovered from autoclaved herring sperm DNA and occurs abundantly as a natural hormone in plants.
ii. Zeatin is a natural cytokinin isolated from corn-kernels and coconut milk.
iii. Cytokinins help produce new leaves, chloroplasts in leaves, and adventitious shoot formation.
iv. Cytokinins promote apical dominance and accelerate leaf senescence.

(A) (i) and (iv) only

(B) (ii) and (iii) only

(C) (i), (ii), and (iii) only

(D) (ii), (iii), and (iv) only

Correct Answer: (B)

Worked Explanation: Statement (i) is incorrect because kinetin does *not* occur naturally in plants. Statement (iv) is incorrect because cytokinins *overcome* apical dominance and *delay* leaf senescence by promoting nutrient mobilisation. Statements (ii) and (iii) are completely accurate.

Practice Question 13 (Single-Choice MCQ)

Which of the following physiological effects is mediated by Ethylene (or its aqueous source Ethephon)?

(A) Speeding up the malting process in the brewing industry

(B) Promoting female flowers in cucumbers and rapid internode/petiole elongation in deep water rice plants

(C) Inducing stomatal closure during drought stress and acting as a general inhibitor of seed germination

(D) Selectively killing dicotyledonous weeds in monocot lawns

Correct Answer: (B)

Worked Explanation: Ethylene promotes rapid internode/petiole elongation in deep water rice plants so leaves/upper shoot parts remain above water. Ethephon (source of ethylene) promotes female flowers in cucumbers, increasing yield. Option (A) is GA_3, Option (C) is ABA, and Option (D) is 2, 4-D (synthetic auxin).

Practice Question 14 (Assertion-Reason MCQ)

Assertion (A): Abscisic acid (ABA) helps seeds withstand desiccation and other factors unfavourable for growth.
Reason (R): ABA acts as a general plant growth inhibitor, inhibits seed germination, and induces seed dormancy.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Explanation: ABA plays a vital role in seed development, maturation, and dormancy. By inhibiting metabolism/germination and inducing dormancy, ABA enables seeds to tolerate desiccation and survive unfavourable environmental conditions.

Practice Question 15 (Single-Choice MCQ)

Which of the following pairs of Plant Growth Regulators act **antagonistically** to each other in regulating seed dormancy and apical dominance, respectively?

(A) Seed dormancy: Auxin vs. Cytokinin; Apical dominance: ABA vs. Gibberellin

(B) Seed dormancy: ABA vs. Gibberellin; Apical dominance: Auxin vs. Cytokinin

(C) Seed dormancy: Ethylene vs. Gibberellin; Apical dominance: Auxin vs. ABA

(D) Seed dormancy: Cytokinin vs. Ethylene; Apical dominance: Gibberellin vs. Auxin

Correct Answer: (B)

Worked Explanation: In most situations, ABA acts as an antagonist to Gibberellins (GAs)—ABA induces seed dormancy while GAs break dormancy and promote germination. In apical dominance, Auxin inhibits lateral bud growth (promotes apical dominance) while Cytokinin promotes lateral shoot growth (overcomes apical dominance).

8. Common Pitfalls & Quick Revision Checklist

High-Yield NEET Pitfalls to Avoid

- **Kinetin vs. Zeatin:** Never mark Kinetin as a naturally occurring plant hormone. Kinetin was isolated from autoclaved herring sperm DNA and **does not occur naturally in plants**. **Zeatin** (from corn kernels and coconut milk) is natural.
- **Auxin's Dual Effect on Abscission:** Auxin prevents fruit and leaf drop at **early stages**, but promotes the abscission of **older mature leaves and fruits**.
- **Flowering in Pineapple:** Both **Auxins** and **Ethylene** promote/initiate flowering in pineapples (Ethylene also synchronises fruit-set in pineapples and induces flowering in mango).
- **Parthenocarpy vs. Fruit Shape:** Auxins induce parthenocarpy in **tomatoes**; Gibberellins elongate and improve the shape of **apples** and increase stalk length in **grapes**; Ethephon hastens ripening in **tomatoes and apples**.
- **2, 4-D Specificity:** 2, 4-D kills **dicotyledonous weeds**, not monocotyledonous lawn grasses.
- **Respiratory Climactic:** The sharp rise in respiration rate during fruit ripening is caused by **Ethylene**, not Gibberellin or Auxin.

Concise Final Recap

- **Growth:** Irreversible permanent increase in size of an organ/part/cell, accompanied by energy-consuming metabolic processes.
- **Meristems & Open Growth:** Primary growth (elongation) via Root and Shoot Apical Meristems; Secondary growth (girth in dicots/gymnosperms) via Vascular Cambium and Cork Cambium.
- **Phases of Growth:** Meristematic (dense protoplasm, thin cellulosic walls, abundant plasmodesmata) → Elongation (vacuolation, new cell wall deposition) → Maturation (maximal wall thickening, protoplasmic modifications).
- **Growth Rates:** Arithmetic ($L_t = L_0 + rt$, linear curve) vs. Geometric ($W_1 = W_0 e^{rt}$, sigmoid S-curve with lag, exponential/log, and stationary phases).
- **Differentiation Cycle:** Meristematic cells → *Differentiation* → Mature cells (e.g., parenchyma) → *Dedifferentiation* → Secondary meristems (interfascicular/cork cambium) → *Redifferentiation* → Secondary permanent tissues.

- **Development & Plasticity:** $\text{Development} = \text{Growth} + \text{Differentiation}$. Plasticity (heterophylly) occurs by life phase (cotton, coriander, larkspur) or environment (buttercup).
- **Plant Growth Regulators:** Promoters (Auxins, Gibberellins, Cytokinins) and Inhibitors (ABA; Ethylene is largely inhibitory but has diverse roles).